

实验二：Serverless Platform

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Step one

1. 初始化模板项目

```
root@ubuntu:~/cloudz# cd /root/cloudz/functions/  
root@ubuntu:~/cloudz/functions# ls  
build template  
root@ubuntu:~/cloudz/functions# export OPENFAAS_URL=http://127.0.0.1:31112  
root@ubuntu:~/cloudz/functions# echo $OPENFAAS_URL  
http://127.0.0.1:31112  
root@ubuntu:~/cloudz/functions# faas-cli login --password admin  
WARNING! Using --password is insecure, consider using: cat ~/faas_pass.txt | faas-cli login -u user  
--password-stdin  
Calling the OpenFaaS server to validate the credentials...  
WARNING! Communication is not secure, please consider using HTTPS. Letsencrypt.org offers free SSL/TLS certificates.  
credentials saved for admin http://127.0.0.1:31112  
root@ubuntu:~/cloudz/functions# fa  
faas-cli      faillog      faked-tcp     fakeroot-sysv fallocate      fatlabel  
factor        faked-sysv    fakeroot      fakeroot-tcp  false  
root@ubuntu:~/cloudz/functions# faas-cli new --lang python3-flask hello-python  
Folder: hello-python created.
```



```
Function created in folder: hello-python  
Stack file written: hello-python.yml  
root@ubuntu:~/cloudz/functions# ls hello-python  
handler.py __init__.py requirements.txt  
root@ubuntu:~/cloudz/functions#
```

2. 修改handler文件并编译运行

```
Successfully built ebc46cb556d3
Successfully tagged hello-python:latest
Image: hello-python:latest built.
[0] < Building hello-python done in 29.18s.
[0] Worker done.

Total build time: 29.18s
root@ubuntu:~/cloudz/functions# faas-cli deploy -f hello-python.yml
Deploying: hello-python.
WARNING! Communication is not secure, please consider using HTTPS. Letsencrypt.org offers free SSL/TLS certificates.

Deployed. 202 Accepted.
URL: http://127.0.0.1:31112/function/hello-python.openfaas-fn

root@ubuntu:~/cloudz/functions# faas-cli list
Function      Invocations    Replicas
hello-python      0              1
root@ubuntu:~/cloudz/functions# faas-cli describe hello-python
Name:          hello-python
Status:        Ready
Replicas:      1
Available replicas: 1
Invocations:   0
Image:         hello-python:latest
Function process: python index.py
URL:           http://127.0.0.1:31112/function/hello-python
Async URL:     http://127.0.0.1:31112/async-function/hello-python
Labels:        faas_function : hello-python
Annotations:   prometheus.io.scrape : false
root@ubuntu:~/cloudz/functions# echo Tom | faas-cli invoke hello-python
Hello Tom
root@ubuntu:~/cloudz/functions# curl http://127.0.0.1:31112/function/hello-python -d "Tom"
Hello Tomroot@ubuntu:~/cloudz/functions# faas-cli list
Function      Invocations    Replicas
hello-python      2              1
root@ubuntu:~/cloudz/functions# A_
```

Step two

float-operation

1. 代码

```
import json
import math
from time import time

def float_operation(N):
    start=time();
    for i in range(0,N):
        sin_i = math.sin(i)
        cos_i = math.cos(i)
        sqrt_i = math.sqrt(i)
    latency = time() - start
    return latency

def handle(req):
    json_req = json.loads(req)
    n = int(json_req['n'])
```

```

latency = float_operation(n)
return "latency: " + str(latency)

```

2. 运行

```

---> Using cache
---> a943dc2bc0fe
Step 27/32 : ENV fprocess="python index.py"
---> Using cache
---> 42cd635e9423
Step 28/32 : ENV cgi_headers="true"
---> Using cache
---> 47867d190d93
Step 29/32 : ENV mode="http"
---> Using cache
---> c0ffada439fd
Step 30/32 : ENV upstream_url="http://127.0.0.1:5000"
---> Using cache
---> 42ddf411b31d
Step 31/32 : HEALTHCHECK --interval=5s CMD [ -e /tmp/.lock ] || exit 1
---> Using cache
---> 3fc3bb382c7d
Step 32/32 : CMD ["fwatchdog"]
---> Using cache
---> 198ac9d5ed10
Successfully built 198ac9d5ed10
Successfully tagged float-operation:latest
Image: float-operation:latest built.
[0] < Building float-operation done in 0.14s.
[0] Worker done.

Total build time: 0.14s
root@ubuntu:~/cloudz/functions# faas-cli deploy -f float-operation.yml
Deploying: float-operation.
WARNING! Communication is not secure, please consider using HTTPS. Letsencrypt.org offers free SSL/TLS certificates.

Deployed. 202 Accepted.
URL: http://127.0.0.1:31112/function/float-operation.openfaas-fn

root@ubuntu:~/cloudz/functions# curl -d '{"n":100}' http://127.0.0.1:31112/function/float-operation
latency: 2.0265579223632812e-05root@ubuntu:~/cloudz/functions#

```

matmul

1. 代码

```

import json
import random
from time import time

def generate_random_matrix(n):
    return [[random.random() for _ in range(n)] for _ in range(n)]

def matmul(A, B):
    n = len(A)
    C = [[0 for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def calculate_latency(n):
    A = generate_random_matrix(n)

```

```

B = generate_random_matrix(n)

start_time = time()
C = matmul(A, B)
latency = time() - start_time
return "latency: " + str(latency)

def handle(req):
    json_req = json.loads(req)
    n = int(json_req['n'])

    res = calculate_latency(n)
    return res

```

2. 运行

```

root@ubuntu:~/cloudz/functions# faas-cli new -lang python3-flask matmul
Folder: matmul created.

Function created in folder: matmul
Stack file written: matmul.yml
root@ubuntu:~/cloudz/functions#

---> Using cache
---> 80a0243c14b0
Step 27/32 : ENV fprocess="python index.py"
---> Using cache
---> 4f49b8520271
Step 28/32 : ENV cgi_headers="true"
---> Using cache
---> 4ca5321cc541
Step 29/32 : ENV mode="http"
---> Using cache
---> 4e2668b1b4ea
Step 30/32 : ENV upstream_url="http://127.0.0.1:5000"
---> Using cache
---> b28be443c4b9
Step 31/32 : HEALTHCHECK --interval=5s CMD [ -e /tmp/.lock ] || exit 1
---> Using cache
---> 7b825e478c4e
Step 32/32 : CMD ["fwatchdog"]
---> Using cache
---> 4ac693e9fcf7
Successfully built 4ac693e9fcf7
Successfully tagged matmul:latest
Image: matmul:latest built.
[0] < Building matmul done in 0.15s.
[0] Worker done.

Total build time: 0.15s
root@ubuntu:~/cloudz/functions# faas-cli deploy -f matmul.yml
Deploying: matmul.
WARNING! Communication is not secure, please consider using HTTPS. Letsencrypt.org offers free SSL/TLS certificates.

Deployed. 202 Accepted.
URL: http://127.0.0.1:31112/function/matmul.openfaas-fn

root@ubuntu:~/cloudz/functions# curl -d '{"n":50}' $OPENFAAS_URL/function/matmul
latency: 0.011615753173828125root@ubuntu:~/cloudz/functions# _

```

资源占用和统计

```

root@ubuntu:~/cloudz/functions# faas-cli deploy -f hello-python.yml
Deploying: hello-python.
WARNING! Communication is not secure, please consider using HTTPS. Letsencrypt.org offers free SSL/TLS certificates.

Deployed. 202 Accepted.
URL: http://127.0.0.1:31112/function/hello-python.openfaas-fn

root@ubuntu:~/cloudz/functions# faas-cli list
Function              Invocations    Replicas
float-operation       9              1
matmul                6              1
hello-python          1              1
root@ubuntu:~/cloudz/functions# docker ps | grep hello-python
3cd315db1281   ebc46cb556d3   "fwatchdog"        21 seconds ago   Up 20 seconds
k8s_hello-python_hello-python-59dcf499b5-mqq62_openfaas-fn_4edd0686-30e2-416c-84a5-a806052e5b8b_0
208c3eb60b6f   rancher/pause:3.1   "/pause"          21 seconds ago   Up 21 seconds
k8s_POD_hello-python-59dcf499b5-mqq62_openfaas-fn_4edd0686-30e2-416c-84a5-a806052e5b8b_0
root@ubuntu:~/cloudz/functions#
CONTAINER ID   NAME
CPU %          MEM USAGE / LIMIT   MEM %           NET I/O         BLOCK I/O        PIDS
d94f0c9017a7   k8s_hello-python_hello-python-5b7c69cc75-67gt8_openfaas-fn_f5f819d7-789e-4543-9432-c9
b776d9d633_0   0.19%               21.76MiB / 7.796GiB   0.27%           0B / 0B         0B / 0B         17

```

使用wrk进行统计

```

root@ubuntu:~/cloudz/functions# wrk -t 1 -c 1 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/matmul
Running 5s test @ http://127.0.0.1:31112/function/matmul
1 threads and 1 connections
  Thread Stats   Avg      Stdev     Max   +/-  Stdev
  Latency    91.25ms    4.49ms  118.66ms   94.55%
  Req/Sec    10.72      4.21    20.00    74.47%
  Latency Distribution
  50%    90.47ms
  75%    92.03ms
  90%    93.55ms
  99%   118.66ms
 55 requests in 5.08s, 10.30KB read
Requests/sec:    10.83
Transfer/sec:     2.03KB
root@ubuntu:~/cloudz/functions# AA

```

Step three

matmul

1. 无限制

```

CONTAINER ID   NAME
CPU %          MEM USAGE / LIMIT   MEM %           NET I/O         BLOCK I/O        PIDS
d94f0c9017a7   k8s_hello-python_hello-python-5b7c69cc75-67gt8_openfaas-fn_f5f819d7-789e-4543-9432-c9
b776d9d633_0   0.19%               21.76MiB / 7.796GiB   0.27%           0B / 0B         0B / 0B         17

```

```

root@ubuntu:~/cloudz/functions# wrk -t 1 -c 1 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/matmul
Running 5s test @ http://127.0.0.1:31112/function/matmul
1 threads and 1 connections
  Thread Stats   Avg      Stdev     Max   +/-  Stdev
  Latency    91.25ms    4.49ms  118.66ms   94.55%
  Req/Sec    10.72      4.21    20.00    74.47%
  Latency Distribution
  50%    90.47ms
  75%    92.03ms
  90%    93.55ms
  99%   118.66ms
 55 requests in 5.08s, 10.30KB read
Requests/sec:    10.83
Transfer/sec:     2.03KB
root@ubuntu:~/cloudz/functions# AA

```

2. 增加资源和请求限制

```

version: 1.0
provider:
  name: openfaas
  gateway: http://127.0.0.1:31112
functions:
  matmul:
    lang: python3-flask
    handler: ./matmul
    image: matmul:latest
    labels:
      com.openfaas.scale.max: 1
      com.openfaas.scale.zero: true
    limits:
      cpu: 100m
      memory: 40Mi
    requests:
      cpu: 100m
      memory: 20Mi

```

```

CONTAINER ID   NAME
CPU %          MEM USAGE / LIMIT   MEM %           NET I/O        BLOCK I/O       PIDS
effe98db034d   k8s_matmul_matmul-f95ff88-1lhgn_openfaas-fn_791ca7e5-f954-46a7-9da1-5a5c4db42558_0
0.16%         21.25MiB / 40MiB    53.13%          0B / 0B        0B / 0B         17

~
root@ubuntu:~/cloudz/functions# wrk -t 1 -c 1 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/matmul
Running 5s test @ http://127.0.0.1:31112/function/matmul
 1 threads and 1 connections
   Thread Stats   Avg      Stdev     Max   +/-  Stdev
   Latency    910.76ms  114.33ms  994.65ms   80.00%
   Req/Sec     0.40      0.55     1.00   60.00%
Latency Distribution
 50%    993.24ms
 75%    993.42ms
 90%    994.65ms
 99%    994.65ms
5 requests in 5.05s, 0.93KB read
Requests/sec:      0.99
Transfer/sec:     187.75B
root@ubuntu:~/cloudz/functions# AA_

```

float-operation

1. 无限制

- 单线程单链接

```

CONTAINER ID   NAME
CPU %          MEM USAGE / LIMIT   MEM %           NET I/O        BLOCK I/O       PIDS
ab06219ce031   k8s_float-operation_float-operation-58d86f8595-swkvxv_openfaas-fn_053b3870-cd47-4644-b
bae-5ba86e4931eb_0
0.05%         25.61MiB / 7.79GiB   0.32%          0B / 0B        0B / 0B         19

^C
root@ubuntu:~/cloudz/functions# wrk -t 1 -c 1 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/float-operation
Running 5s test @ http://127.0.0.1:31112/function/float-operation
 1 threads and 1 connections
   Thread Stats   Avg      Stdev     Max   +/-  Stdev
   Latency     1.77ms    1.47ms   31.41ms   97.85%
   Req/Sec    603.12     53.02   690.00    80.00%
Latency Distribution
 50%     1.58ms
 75%     1.81ms
 90%     2.16ms
 99%     5.30ms
3007 requests in 5.01s, 571.96KB read
Requests/sec:     600.08
Transfer/sec:     114.14KB

```

- 单线程5条链接

```
root@ubuntu:~/cloudz/functions# wrk -t 1 -c 5 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/float-operation
Running 5s test @ http://127.0.0.1:31112/function/float-operation
 1 threads and 5 connections
  Thread Stats   Avg      Stdev     Max   +/-  Stdev
    Latency    6.14ms    2.86ms   32.95ms   80.50%
   Req/Sec    836.88    106.07    1.18k    82.00%
  Latency Distribution
    50%    5.56ms
    75%    7.10ms
    90%    9.37ms
    99%   17.03ms
 4170 requests in 5.00s, 792.50KB read
Requests/sec:   833.18
Transfer/sec:   158.35KB
root@ubuntu:~/cloudz/functions#
```

2. 限制资源

```
version: 1.0
provider:
  name: openfaas
  gateway: http://127.0.0.1:31112
functions:
  float-operation:
    lang: python3-flask
    handler: ./float-operation
    image: float-operation:latest
    labels:
      com.openfaas.scale.max: 1
      com.openfaas.scale.zero: true
    limits:
      cpu: 100m
      memory: 40Mi
    requests:
      cpu: 100m
      memory: 20Mi
```

- 单线程单链接

```
CONTAINER ID   NAME
fb6292b3cdeb   k8s_float-operation_float-operation-7f75cd4f99-71smz_openfaas-fn_33f49cd9-a334-434f-9
c05-dd7e055e95b6_0   0.01%    23.32MiB / 40MiB    58.31%    0B / 0B    0B / 0B    21
^C
root@ubuntu:~/cloudz/functions# wrk -t 1 -c 1 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/float-operation
Running 5s test @ http://127.0.0.1:31112/function/float-operation
 1 threads and 1 connections
  Thread Stats   Avg      Stdev     Max   +/-  Stdev
    Latency   27.58ms   31.38ms   98.46ms   77.36%
   Req/Sec    75.02    21.70    118.00    70.00%
  Latency Distribution
    50%    2.43ms
    75%   52.57ms
    90%   78.01ms
    99%   94.68ms
 378 requests in 5.06s, 71.77KB read
Requests/sec:   74.65
Transfer/sec:   14.17KB
```

- 单线程5条链接

```
root@ubuntu:~/cloudz/functions# wrk -t 1 -c 5 -d 5s --latency -s post.lua http://$OPENFAAS_URL/function/float-operation
Running 5s test @ http://127.0.0.1:31112/function/float-operation
 1 threads and 5 connections
  Thread Stats   Avg      Stdev     Max   +/-  Stdev
    Latency    91.33ms   66.01ms  308.56ms   58.14%
    Req/Sec    53.81     23.46   110.00    59.46%
  Latency Distribution
    50%    97.80ms
    75%   103.43ms
    90%   194.48ms
    99%   301.03ms
 212 requests in 5.10s, 40.26KB read
Requests/sec:    41.59
Transfer/sec:     7.90KB
root@ubuntu:~/cloudz/functions#
```

结果分析

- **线程和连接的影响：**增加连接数（从1到5）显著增加了平均延迟，但请求速率却降低了。这表明在保持单个线程的情况下，增加连接数对性能有负面影响。
- **延迟分布：**增加连接数后，延迟分布的各个百分位数都显著增加，表明更多的请求需要更长的时间来完成。
- **数据传输：**尽管请求速率降低，但每秒读取的数据量增加了（从71.77KB到110.00KB），这可能是由于更多的连接导致更多的数据被传输。

总体来看，这些结果表明在单线程情况下，增加连接数对性能有负面影响，特别是在延迟方面。这可能是由于服务器资源（如CPU、内存或网络带宽）的限制导致的。